Project: Worked examples
Title: Enclosure heat dissipation and fan sizing
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Total heat load inside a control enclosure, the effective surface that can dissipate it for the mounting condition, the internal temperature rise with natural convection and radiation through a stated overall heat-transfer coefficient, the forced-ventilation airflow for a target rise (density corrected for altitude), the fan duty point against a fan curve and the system resistance, optional solar gain, and the cooling-unit capacity needed when the limit cannot be met.

WHAT IT CALCULATES	RELATION
Heat dissipated inside the enclosure, plus any solar gain	$P = \Sigma P_i n_i d_i (+ \alpha I A_{sun})$
Effective cooling surface, each face weighted by how freely it sheds heat	$A_{eff} = \Sigma f_{face} \times A_{face}$
Temperature rise with natural cooling, one coefficient over the exposed faces (the IEC TR 60890 tables are not used)	$\Delta T = P/(h A_{eff})$
Air flow a fan must move to hold the target rise	$V = P/(\rho c_p \Delta T_{target})$, with $\rho = p(z)/(287.055 T)$ and c_p = 1006 J/(kg·K)
Fan duty point, where the fan curve meets the system resistance (found by bisection)	$p_{fan}(V) = p_{sys} (V/V_{ref})^2$
Cooling unit capacity, when the surfaces alone cannot reject the heat	$P_{cool} = P - h A_{eff} (T_{max} - T_{amb})$
Sealed enclosure (sealing temperature T_{seal} entered, no breather), pressure difference across the walls when the air warms or cools from the sealing temperature at constant volume, in absolute temperatures (a rise when $T_{int} > T_{seal}$, a suction when below)	$\Delta p = p_{site} \cdot \lvert T_{int}/T_{seal} - 1 \rvert$
A pressure-compensation element (breather or relief) limits it to its cracking pressure	$dp_{seal} = \min(\Delta p, dp_{relief})$
Force of that pressure on a cover of area A_{cover}	$F_{seal} = dp_{seal} \cdot A_{cover}$

Basis: A method of temperature-rise verification of low-voltage switchgear and controlgear assemblies by calculation[1]; Effective surface, heat-transfer coefficient and fan-sizing practice (cited, not copied)[2]; Verification of temperature rise[3](10.10).

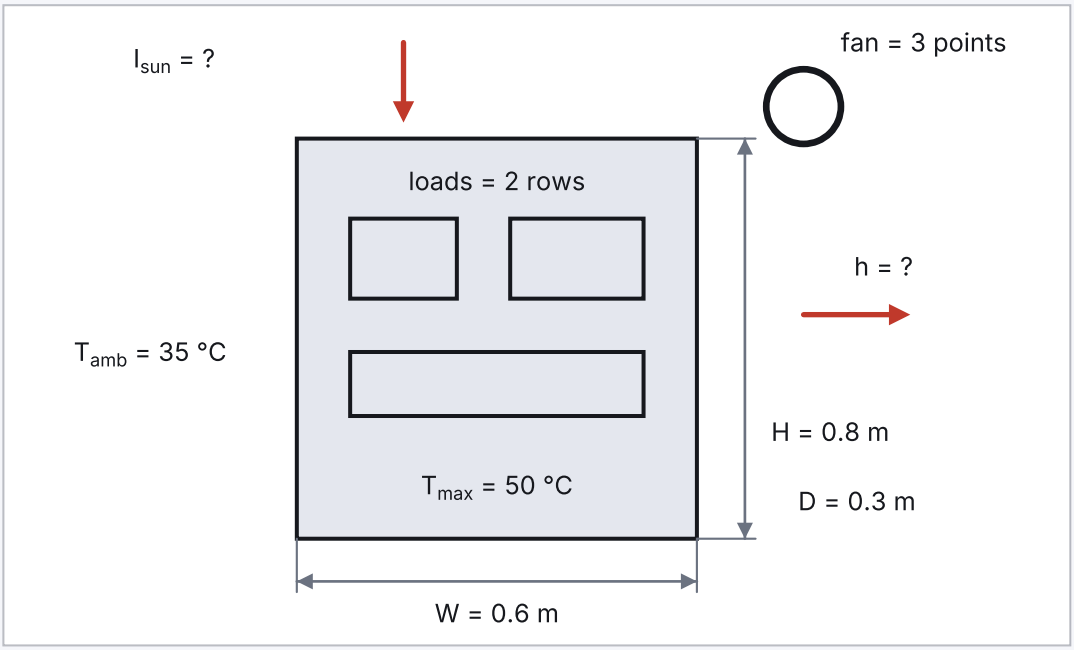


Figure 1 — An enclosure W by H by D holding the heat loads as listed, losing heat through its exposed faces to the ambient, with a fan and the sun on it. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
loads	Heat loads	2 row(s)		spec: Panel schedule PNL-04	any
P_{add}	Additional heat load (linked from E6's dissipated heat or another calculation)	0	W	default	0 … 100000 W
I_{sun}	Solar irradiance on the enclosure (0 = indoors)	0	W/m²	default	0 … 1200 W/m²
A_{sun}	Sunlit area	0	m²	default	0 … 100 m²
α_{sun}	Surface absorptivity	0.5		default	0.1 … 1
W	Enclosure width	0.6	m	spec: Enclosure datasheet AE-1380: 600 × 800 × 300	0.1 … 5 m
H	Enclosure height	0.8	m	spec: Enclosure datasheet AE-1380: 600 × 800 × 300	0.1 … 5 m
D	Enclosure depth	0.3	m	spec: Enclosure datasheet AE-1380: 600 × 800 × 300	0.1 … 5 m
f_{top}	Top exposure (1 free, 0 covered)	1		default	0 … 1
f_{front}	Front exposure	1		default	0 … 1
f_{rear}	Rear exposure (0 = wall-mounted)	0		default	0 … 1

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
f_{left}	Left side exposure	1		default	0 … 1
f_{right}	Right side exposure	1		default	0 … 1
f_{bottom}	Bottom exposure (0 = on the floor)	0		default	0 … 1
h	Overall heat-transfer coefficient of the enclosure surface (editable reference value)	5.5	W/(m²·K)	default	3 … 8 W/(m²·K)
T_{amb}	Ambient temperature	35	°C	spec: Site conditions SC-2: 35 °C plant room; 50 °C inside limit for the drives	-20 … 55 °C
T_{max}	Maximum internal temperature	50	°C	spec: Site conditions SC-2: 35 °C plant room; 50 °C inside limit for the drives	20 … 80 °C
altitude	Site altitude (air density)	0	m	assumption: Wall mounted on a floor plinth: rear and bottom covered; indoors.	0 … 4000 m
V_{pts}	Fan curve: airflow points	[0, 100, 200]	m³/h	spec: Fan datasheet FF-120: curve points	any
p_{pts}	Fan curve: static pressure points	[60, 30, 0]	Pa	spec: Fan datasheet FF-120: curve points	any
p_{sys}	System resistance (filters, grilles) at the reference airflow	20	Pa	spec: Filter datasheet: 20 Pa at 100 m³/h	0 … 500 Pa
V_{ref}	Reference airflow for the system resistance	100	m³/h	spec: Filter datasheet: 20 Pa at 100 m³/h	1 … 100000 m³/h

TABLE ROW	VALUES
loads[1]	name = PLC and I/O; P = 80 W; qty = 1; duty = 1
loads[2]	name = drives; P = 55 W; qty = 4; duty = 1

3. Assumptions

A1. Heat leaves the enclosure through its exposed faces at one overall coefficient h (natural convection plus radiation to surroundings at the ambient temperature); covered faces dissipate nothing. (IEC TR 60890-style effective-surface approach; h is a reference value the user may edit.)
A2. Forced ventilation removes the full heat load with air entering at ambient and leaving at the target rise; natural dissipation is neglected for the airflow (conservative). (Manufacturer fan-sizing practice.)
A3. A sealed enclosure (T_{seal} entered) is a closed, rigid, leak-free volume of dry air at the site pressure when closed: the pressure difference follows from the ideal-gas relation at constant volume, from the sealing temperature to the internal temperature with natural dissipation T_{int} (the temperature without a

cooling unit); a breather, a seal that leaks or a flexing wall reduces it, so it is an upper bound (a pressure-compensation element is modelled only as a cap at its cracking pressure, with no flow resistance); the site pressure is the standard atmosphere at the altitude. (Ideal gas at constant volume (derived).)
A4. The fan curve is linear between the entered points; the system resistance scales with airflow squared. (Duty-point idealisation.)
A5. altitude (Site altitude (air density)) is taken as 0 m. (Wall mounted on a floor plinth: rear and bottom covered; indoors.)

4. Working

[18] $P_{loads} = \Sigma\ P_i \times n_i \times d_i$
$= 2\ \text{item(s)}$
$= 300\ \text{W}$
Internal heat load

[19] $P_{solar} = \alpha_{sun} * I_{sun} * A_{sun}$
$= 0.5 \times 0\ \text{W/m}^2 \times 0\ \text{m}^2$
$= 0\ \text{W}$
Solar gain

[20] $P_{total} = P_{loads} + P_{add} + P_{solar}$
$= 300\ \text{W} + 0\ \text{W} + 0\ \text{W}$
$= 300\ \text{W}$
Total heat to remove

[21] $A_{eff} = f_{top} * W * D + f_{bottom} * W * D + f_{front} * W * H + f_{rear} * W * H + f_{left} * H * D + f_{right} * D$ [1](approach)
$= 1 \times 0.6\ \text{m} \times 0.3\ \text{m} + 0 \times 0.6\ \text{m} \times 0.3\ \text{m} + 1 \times 0.6\ \text{m} \times 0.8\ \text{m} + 0 \times 0.6\ \text{m} \times 0.8\ \text{m} + 1 \times 0.8\ \text{m} \times 0.8\ \text{m} \times 0.3\ \text{m}$
$= 1.14\ \text{m}^2$
Effective dissipating surface

[22] $dT_{nat} = P_{total} / (h * A_{eff})$
$= 300\ \text{W} / (5.5\ \text{W/(m}^2\cdot\text{K)} \times 1.14\ \text{m}^2)$
$= 47.847\ \text{K}$
Temperature rise with natural dissipation

[23] $T_{int} = T_{amb} + dT_{nat}$
$= 35\ \text{degC} + 47.847\ \text{K}$
$= 82.847\ \text{degC}$
Internal temperature with natural dissipation

[24] $dT_{allow} = T_{max} - T_{amb}$
$= 50\ \text{degC} - 35\ \text{degC}$
$= 15\ \text{K}$
Allowed rise

[25] $u_{nat} = dT_{nat} / dT_{allow}$
$= 47.847\ \text{K} / 15\ \text{K}$
$= 3.1898$
Natural dissipation utilisation

[26] $P_{nat,max} = h * A_{eff} * dT_{allow}$
$= 5.5\ \text{W/(m}^2\cdot\text{K)} \times 1.14\ \text{m}^2 \times 15\ \text{K}$
$= 94.05\ \text{W}$
Heat the enclosure dissipates naturally at the limit

[27] $P_{cool} = \max(0\ \text{W}, P_{total} - P_{nat,max})$
$= \max(0\ \text{W}, 300\ \text{W} - 94.05\ \text{W})$
$= 205.95\ \text{W}$
Cooling-unit capacity needed if ventilation is not used

[28] p_{site} = 101325 (1 - 2.25577e-5 z)^5.2559
= z = 0 m
= 101.32 kPa
Site pressure
[29] ρ_{air} = p_{site} / (287.055 J/(kg*K) * T_{amb})
= 101.32 kPa / (287.055 J/(kg*K) × 35 degC)
= 1.1455 kg/m³
Air density at ambient and altitude
[30] V_{req} = P_{total} / (ρ_{air} * 1006 J/(kg*K) * dT_{allow})
= 300 W / (1.1455 kg/m³ × 1006 J/(kg*K) × 15 K)
= 62.481 m³/h
Airflow for forced ventilation at the allowed rise (natural dissipation neglected)
[31] $V_{req,net}$ = max(0 m^3/h, (P_{total} - $P_{nat,max}$) / (ρ_{air} * 1006 J/(kg*K) * dT_{allow}))
= max(0 m^3/h, (300 W - 94.05 W) / (1.1455 kg/m³ × 1006 J/(kg*K) × 15 K))
= 42.893 m³/h
Airflow crediting natural dissipation
[36] V_{duty} = $p_{fan}(V)$ = p_{sys} $(V/V_{ref})^2$, bisection
=
= 113.75 m³/h
Fan duty airflow
[37] p_{duty} = p_{sys} $(V_{duty}/V_{ref})^2$
= 20 × (113.7/100)²
= 25.876 Pa
Static pressure at duty
[38] u_{fan} = V_{req} / V_{duty}
= 62.481 m³/h / 113.75 m³/h
= 0.5493
Airflow utilisation (required / delivered)

5. Results

SYMBOL	VALUE	DESCRIPTION
P_{loads}	300 W	Internal heat load
P_{solar}	0 W	Solar gain
P_{total}	300 W	Total heat
A_{eff}	1.14 m²	Effective surface
dT_{nat}	47.847 K	Natural temperature rise
T_{int}	82.847 degC	Internal temperature (natural)
$P_{nat,max}$	94.05 W	Natural dissipation at the limit
P_{cool}	205.95 W	Cooling unit needed
T_{max}	50 degC	Internal temperature limit
ρ_{air}	1.1455 kg/m³	Air density
V_{req}	62.481 m³/h	Required airflow
$V_{req,net}$	42.893 m³/h	Required airflow (net)
V_{duty}	113.75 m³/h	Fan duty airflow
p_{duty}	25.876 Pa	Fan duty pressure
u_{fan}	0.5493	Airflow utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Natural dissipation holds T_{max}	$u_{nat} = 3.1898$	$u_{nat} \leq 1$	3.19	
PASS	Fan delivers the required airflow	$u_{fan} = 0.5493$	$u_{fan} \leq 1$	0.549	

7. Applicability

WARNING — input.P_add: P_{add} (Additional heat load (linked from E6's dissipated heat or another calculation)) was not entered; the template default 0 W is used.
WARNING — input.I_sun: I_{sun} (Solar irradiance on the enclosure (0 = indoors)) was not entered; the template default 0 W/m^2 is used.
WARNING — input.A_sun: A_{sun} (Sunlit area) was not entered; the template default 0 m^2 is used.
WARNING — input.alpha_sun: α_{sun} (Surface absorptivity) was not entered; the template default 0.5 is used.
WARNING — input.f_top: f_{top} (Top exposure (1 free, 0 covered)) was not entered; the template default 1 is used.
WARNING — input.f_front: f_{front} (Front exposure) was not entered; the template default 1 is used.
WARNING — input.f_rear: f_{rear} (Rear exposure (0 = wall-mounted)) was not entered; the template default 0 is used.
WARNING — input.f_left: f_{left} (Left side exposure) was not entered; the template default 1 is used.
WARNING — input.f_right: f_{right} (Right side exposure) was not entered; the template default 1 is used.
WARNING — input.f_bottom: f_{bottom} (Bottom exposure (0 = on the floor)) was not entered; the template default 0 is used.
WARNING — input.h: h (Overall heat-transfer coefficient of the enclosure surface (editable reference value)) was not entered; the template default 5.5 W/(m^2*K) is used.
WARNING — cooling-unit: T_{max}: natural dissipation holds 94.05 W of 300 W; ventilate or fit a 206 W cooling unit.

8. Verification

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Template elec.enclosure-heat@1.5.0 Content hash 857f14f33a5fb9c0 Engine calc-engine/1.0.0
Known-answer cases passed 4/4 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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9. References

1. International Electrotechnical Commission. A Method of Temperature-Rise Verification of Low-Voltage Switchgear and Controlgear Assemblies by Calculation. IEC TR 60890. IEC.
2. Rittal; nVent Hoffman. Enclosure climate control guides [manufacturer technical literature].
3. British Standards Institution. Low-Voltage Switchgear and Controlgear Assemblies—Part 1: General Rules. BS EN 61439-1. BSI.